

Homework 6

Jizhou Yan ^{TA}

Instructions before the assignment 作业前须知:

- This assignment is for the **seventh week's homework** of the System Dynamics and Control course (hereinafter referred to as SDC). The submission deadline is **before 11:59 PM on October 31, 2024**. Please ensure timely submission, as late submissions will result in point deductions. 本次作业为 SDC 第七周课程作业, 提交时间为 **2024/10/31 23:59** 前。
 - If submitted within 12 hours after the deadline, no points will be deducted (**this is allowed only twice, so focus on completing it properly rather than rushing to meet the deadline if you're unable to finish on time**);
 - If submitted within three days after the deadline, **20% of the total score of this assignment will be deducted**;
 - If submitted within two week after the deadline, **30% of the total score of this assignment will be deducted**;
 - Any other late submissions will result in a **40% deduction**;
- For late submissions, please contact the teaching assistant(TA), **Yan Jizhou**, via email at **yanjizhou312@163.com** or WeChat **y3139838452**. Make sure to use the filename format as "SDC-Assignment-x-Resubmission-Name-Student ID." 作业补交时通过邮箱或微信与助教闫济洲联系, 邮件标题及文件名备注为 "SDC-第 x 次作业补交-姓名-学号".
- The assignment is graded out of 10 points. It consists of basic course questions (10 points) and extension questions (0 to 5 points). **The final assignment score will be the smaller one between 10 and the actual score.** Each assignment earns all points by completing only the basic assignment. 每次作业仅完成基础作业即可获得所有分数, 鼓励同学们完成拓展课程题目。
- Students are encouraged to attempt the extension questions. Any extra points earned will be proportionally added to the final course grade for possible grade improvement.
- **Academic misconduct such as plagiarism is strictly prohibited in course work. Plagiarism and other academic misconduct will be dealt with in accordance with the relevant regulations of the University.** If you have discussed the assignment with classmates or received help, please note at the beginning of the assignment: "This assignment was completed after discussion with XXX (classmate's name), and the thought process was referenced from XXX (classmate's name)." **Every contribution deserves recognition. Unacknowledged similarities between assignments will be treated as plagiarism.** 课程作业中严禁抄袭等学术不端行为。如作业完成过程中经历了同学讨论/同学帮助的情况, 请在作业最前面注明 "本次作业经过与 XXX (同学姓名) 讨论, 参考了 XXX (同学姓名) 的部分思路"。每位同学的帮助都值得被记录。不被注明的作业雷同将被认为是抄袭行为。

1. Problem 1

Find static position error coefficients, static velocity error coefficients, static acceleration error coefficients and steady state errors of unit feedback systems with the open-loop transfer function as follows. Suppose the input signals are unit step, unit ramp and unit acceleration respectively.

$$(1) G(s) = \frac{50}{(0.1s+1)(2s+1)} \quad (2) G(s) = \frac{K(2s+1)(4s+1)}{s^2(s^2+2s+10)}$$

2. Problem 2

Consider the second order differential equation given below,

$$M\ddot{x} + B\dot{x} + Kx = f(t)$$

with $M = 5$ kg, $B = 1$ N · s/m, and $K = 100$ N/m. (A) Find the steady-state response, $x_{ss}(t)$, of the system to a co-sinusoidal input,

$$f(t) = (2 \cos(2t + 1.5)) \text{ Newtons}$$

(B) Find the steady-state response, $x_{ss}(t)$, of the system to an input,

$$f(t) = (5 \sin(4t + 0.5) + 10 \cos(2t) + 2) \text{ Newtons}$$

(C) For a range of frequencies 0rad/s to 15rad/s in steps of 0.1rad/s, find and plot the ratio of the steady-state sinusoidal response amplitude to the input amplitude. Also find and plot the relative phase between the steady-state sinusoidal response and the input.

(D) By rewriting the equation in the form,

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x = \frac{f(t)}{M}$$

please prove that the ratio between the steady-state sinusoidal response amplitude and the input sinusoidal amplitude is a maximum when the input frequency is,

$$\omega_{\max} = \sqrt{1 - 2\zeta^2}\omega_n$$

and that the maximum ratio is,

$$\left(\frac{x_{ss}}{f_{ss}}\right)_{\max} = \frac{1/K}{2\zeta\sqrt{1 - \zeta^2}}$$